Macro-systemic Approach to Multi-Asset Investing

Risk Profile Analysis: Gold (GLD)

March 25, 2026



Alpha Capital Jr.
Buy-Side

Author:
Nicolas Poncin

Executive summary

This document is part of a macro-driven multi-asset allocation research project. The objective is to design a rules-based framework that translates macroeconomic regimes into systematic portfolio allocations and decisions across a set of assets, targeting risk-adjusted alpha over equity benchmarks.

This section covers individual asset analytics: return distribution, performance, risk, volatility, Sharpe, Sortino, VaR, CVaR, maximum drawdown, and time under water. Each asset is analyzed independently before being integrated into the broader cross-asset and allocation framework.

This report focuses on Gold (GLD). Its role in the portfolio is to hedge against inflation and stress periods.

Assets choice

Class	Product	Ticker	First Obs.
Equity	ETF S&P 500	SPY	29-01-1993
Commodities	SPDR Gold Shares	GLD	18-11-2004
Commodities	Commodity Index Tracking Fund	DBC	06-02-2006
Bonds	Treasuries 20Y+	TLT	30-07-2002
Credit*	iShares Corp Bond	LQD	TBD
Common period			2006-02-06

Table: Selected assets for a multi-asset portfolio

*To be implemented after further research on the credit cycle and its impact on the economy.

Gold - General Description

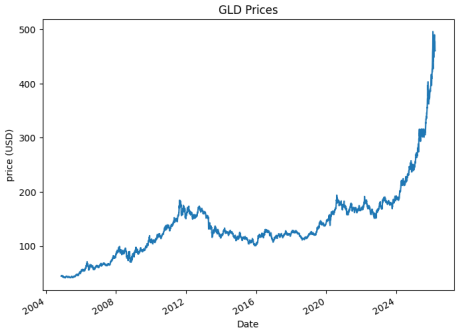
Characteristics:

- **Extreme simplicity** – single asset class
  - Holdings: 100% physical gold (monolithic position)
- **Largest** & **highly liquid** physical gold ETF (grantor trust structure)
  - Holds 99.5%+ purity gold bullion as London Good Delivery bars
  - $150B+ AUM
- Expensive ETFs but lower cost than spot gold
  - 0.40% expense ratio for storage, insurance, custody
  - 4.2x more expensive than SPY, creating long-term drag
- Alternatives: Cheaper IAU or GLDM

→ Gold is a **macro hedge**, not a return generator.

Gold - Historic performances

The **SPDR Gold Shares** serves as a macro hedge in the portfolio, designed to provide diversification across asset classes.



Performance:

- **Period:** 2004–2026 (22 years)
- **CAGR:** 10.91%
- **Annualized Volatility:** 18.05%
- **Sharpe Ratio:** 0.580
- **Sortino Ratio:** 0.817

Insight: Despite lower CAGR than the equity class (S&P500), GLD exhibits better risk-adjusted returns. The precious metal surged 64% in 2025, outperforming all major US indices. $\rightarrow$ Sortino $>$ Sharpe indicates left-tail protection dominates; useful as portfolio ballast.

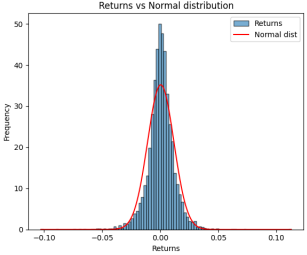
Gold - Distribution of returns

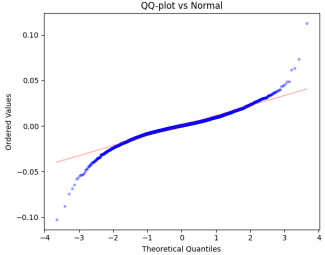
Distribution parameters

Metric	Value
Mean	0.048%
Skewness	-0.318
Kurtosis	6.810
Exc. Kurtosis	3.810

Risk metrics

CL	VaR	CVaR
95%	1.79%	2.73%
97.5%	2.35%	3.44%
99%	3.18%	4.53%





- **Slight left-tail bias** from negative skew: marginally larger extreme losses than gains.
- **Fat tails** present but moderate: extreme moves occur more often than normal, but less severe than SPY (kurtosis 8.86).
- **QQ-plot** S-curve confirms clear deviation at both tails; Gaussian VaR underestimates tail risk.

Insights:

→ Critically, GLD's left tail is *decorrelated* from SPY's. During equity crashes, gold rises or stays flat. Their left tails don't coincide, making it an effective portfolio hedge despite modest negative skew.

Gold - Drawdowns analysis

Mini-Correction (2006)
Rapid correction, slow recovery

Metric	Value
Peak	May 12, 2006
Trough	Jun 14, 2006
Recovery	Sep 18, 2007
Max Drawdown	-21.79%
TUW	339d
Recovery Time	317d

Global Financial Crisis (2008)

Metric	Value
Peak	Mar 17, 2008
Trough	Nov 12, 2008
Recovery	Sep 16, 2009
Max Drawdown	-29.41%
TUW	379d
Recovery Time	211d

Secular Bear Market (2011–2020)
Largest drawdown: equity bull market

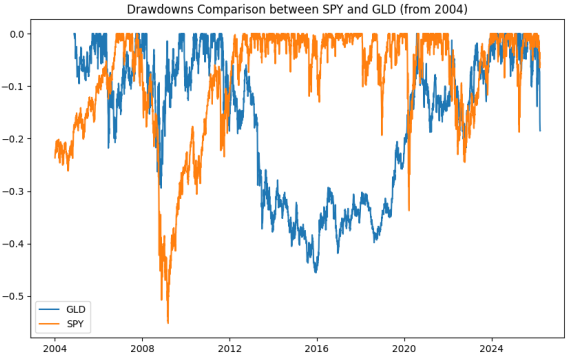
Metric	Value
Peak	Aug 22, 2011
Trough	Dec 17, 2015
Recovery	Jul 29, 2020
Max Drawdown	**-45.56%**
TUW	**2,248d**
Recovery Time	**1,160d**

Rate Hike Cycle (2020–2024)
Period of high interest rates

Metric	Value
Peak	Aug 6, 2020
Trough	Sep 26, 2022
Recovery	Mar 4, 2024
Max Drawdown	-22.00%
TUW	898d
Recovery Time	360d

Gold - Comparison with SPY

Historical drawdowns:



2008 GFC: GLD vs SPY

Metric	GLD	SPY
Peak	Mar 17, 2008	Oct 9, 2007
Trough	Nov 12, 2008	Mar 9, 2009
Recovery	Sep 16, 2009	Aug 16, 2012
Max Drawdown	-29.41%	-55.19%
TUW	379d	1,224d
Recovery Time	211d	869d

In 2008, gold absorbed the shock with half the drawdown and recovered 4x faster than SPY, confirming its role as a macro hedge during systemic crises.

Gold - Quick comparison with TLT

2008 GFC — Crisis Regime

Metric	GLD	TLT	SPY
Peak	Mar 17, 2008	Dec 30, 2008	Oct 9, 2007
Trough	Nov 12, 2008	Jun 10, 2009	Mar 9, 2009
Recovery	Sep 16, 2009	Aug 18, 2011	Aug 16, 2012
Max DD	-29.41%	-26.59%	-55.19%
TUW	379d	664d	1,224d
Recovery	**211d**	553d	869d

GLD recovers 2.6x faster than TLT

2020–2024 Rate Hike Cycle

Metric	GLD	TLT
Peak	Aug 6, 2020	Aug 4, 2020
Trough	Sep 26, 2022	Oct 19, 2023
Recovery	Mar 4, 2024	Not recovered
Max DD	**-22.00%**	-48.35%
TUW	898d	1,415d+
Recovery	360d	N/A

TLT crushed by duration risk; GLD fully recovered

Conclusion:

→ Gold proves to be a more regime-agnostic hedge than TLT. While long-duration bonds outperform in deflationary crises characterized by falling rates (2008, 2011), they are severely exposed to rate hike cycles (-48% since 2020 and not recovered yet).

→ During the 2008 crisis, gold first declined due to forced institutional deleveraging (Bear Stearns collapse, margin calls). During this time, TLT surged +33.95% as investors fled to Treasuries. Once liquidity pressures eased, gold repriced as a store of value and recovered in 7 months, well ahead of both equities and bonds. Bonds subsequently reversed as markets priced in recovery and rising rates.

Alpha Capital Jr.

Buy-Side

Macro-systemic Approach to multi-asset investing

Nicolas Poncin is a M.Sc. Financial Engineering student with a strong interest in global macroeconomics and multi-asset portfolio management. He aims to apply his academic background to real investment decision-making. This project was developed as a practical research initiative to explore how macroeconomic regimes influence asset behaviour and portfolio allocation, intending to build a disciplined and testable investment process.



Author: Nicolas Poncin

CV — LinkedIn — GitHub — Strava — Instagram